The Assignment Problem

What an Assignment problem is?

- When we want to solve a linear programming problem with special characteristic such as a Square Matrix (i.e. No. of destinations aresame as no. of sources)
- When each source (resource facility) can be associated with one and only one job/destination (One job to one machine)
- Association is to be done in such a way to maximize or minimize the total effectiveness.

So, an assignment problem may be considered as a special type of transportation problem in which

- the capacity of each of the sources as well as the requirement of each of the destinations is taken as 1.
- the given matrix must necessarily be a square matrix,

(which is not the condition for a transportation problem).

				1
				1
				1
1	1	1	1	1

The two type of the assignment problem

Balanced problem

number of rows is equal to the number of columns is said to be an balanced problem.

Unbalanced problem

The number of rows is not equal to the number of columns is said to be an unbalanced problem. Such a problem is handled by introducing dummy row(s) if the number of rows is less than the number of columns and dummy column(s) if the number of columns is less than the number of rows. All elements for such a dummy row/column are taken as zero.

The Assignment Problem

- In many business situations, management needs to assign - personnel to jobs, - jobs to machines, - machines to job locations, or - salespersons to territories.
- Consider the situation of assigning *n* jobs to *n* machines.
- When a job i (=1,2,....,n) is assigned to machine j (=1,2, ...n) that incurs a cost Cij.
- The objective is to assign the jobs to machines at the least possible total cost.

Mathematical formulation (known as the cost matrix)

		Machine				
		1	2	....	n	Source
	1	C11	C12	....	C1n	1
	2	C21	C22	....	C2n	1
Job	.	.	.		.	.
	.	.	.		.	.
	.	.	.		.	.
	n	Cn1	Cn2	....	Cnn	1
Destination		1	1	......	1	

where c_{ij} represents the amount of time taken by i^{th} person to complete the j^{th} job. Let x_{ij} denote the j^{th}job assigned to the i^{th} person. Then, mathematically, the assignment problem can be stated as follows:

Mathematical formulation (known as the cost matrix)

	Machine					
		1	2	....	n	Source
Job	1	C11	C12	....	C1n	1
	2	C21	C22	....	C2n	1
	.	.	.		.	.
	.	.	.		.	.
	.	.	.		.	.
	n	Cn1	Cn2	....	Cnn	1
Destination		1	1	......	1	

The assignment model can be expressed mathematically as follows:

Xij= 0, if the job j is not assigned to machine i

1, if the job j is assigned to machine i

The mathematical model of AP can be formulated as given below:

Minimize $\sum_{i=1}^{n}\sum_{j=1}^{n} x_{ij}c_{ij}$, where i $1, 2, \dots, n$ and j $1, 2, \dots, n$.

Subject to the constraints,

$$x_{ij} = \begin{cases} 1, i^{th} person\ is\ assigned\ the\ j^{th}\ job \\ 0, i^{th} person\ is\ not\ assigned\ the\ j^{th}\ job \end{cases}$$

$x_{i1}+x_{i2}+\dots\dots+x_{in}=1$, i= $1, 2, \dots, n$ ((one job is done by the i^{th} person)

$x_{1j}+x_{2j}+\dots\dots+x_{nj}=1$, j= $1, 2, \dots, n$ (only one person is assigned the j^{th} job)

The constant cij in the above problem represents time. It may be cost or some other parameter which is to be minimised in the assignment problem under consideration.

The Assignment Problem Example

- Assigning sales man to sales territory
- Vehicles and drivers to different routes
- Product to factories
- Jobs to a machines
- Contract to Bidders
- Research problem to teams
- Can you think of Problem that suits to Assignment
- methodology?

The Assignment Problem Example

- Ballston Electronics manufactures small electrical devices.
- Products are manufactured on five different assembly lines (1,2,3,4,5).
- When manufacturing is finished, products are transported from the assembly lines to one of the five different inspection areas (A,B,C,D,E).
- Transporting products from five assembly lines to five inspection areas requires different times (in minutes)

The Assignment Problem Example

	Inspection Area				
Assembly Line	A	B	C	D	E
1	10	4	6	10	12
2	11	7	7	9	14
3	13	8	12	14	15
4	14	16	13	17	17
5	19	11	17	20	19

Under current arrangement, assignment of inspection areas to the assembly lines are 1 to A, 2 to B, 3 to C, 4 to D, and 5 to E.

This arrangement requires 10+7+12+17+19 = 65 man minutes.

Let us explain the Hungarian method of finding the optimal solution of an assignment problem

The steps for obtaining an optimal solution of an assignment problem are as follows:

1. Check whether the given matrix is square. If not, make it square by adding a suitable number of dummy rows (or columns) with 0 cost/time elements.
2. Locate the smallest cost element in each row of the cost matrix. Subtract the smallest element of each row from every element of that row.
3. In the resulting cost matrix, locate the smallest element in each column and subtract the smallest element of each column from every element of that column.

4. In the resulting matrix, search for an optimum assignment as follows:

i) Examine the rows successively until a row with exactly one zero is found. Draw a rectangle around this zero (as 0) and cross out all other zeroes in the corresponding column. Proceed in this manner until all the rows have been examined. If there is more than one zero in any row, do not touch that row; pass on to the next row.

ii) Repeat step (i) above for the columns of the resulting cost matrix.

iii) If a row or column of the reduced matrix contains more than one zeroes, arbitrarily choose a row or column having the minimum number of zeroes. Arbitrarily select any zero in the row or column so chosen. Draw a rectangle around it and cross out all the zeroes in the corresponding row and column. Repeat steps (i), (ii) and (iii) until all the zeroes have either been assigned (by drawing a rectangle around them) or crossed.

iv) If each row and each column of the resulting matrix has one and only one assigned 0, the optimum assignment is made in the cells corresponding to **0** . The optimum solution of the problem is attained and you can stop here.

Otherwise, go to the next step.

5. Draw the minimum number of horizontal and/or vertical lines through all the zeroes as follows:

i) Tick mark (√) the rows in which assignment has not been made.

ii) Tick mark (√) columns, which have zeroes in the marked rows.

iii) Tick mark (√) rows (not already marked) which have assignments in marked columns. Then tick mark (√) columns, which have zeroes in newly marked rows, if any. Tick mark (√) rows (not already marked), which have assignments in these newly marked columns.

iv) Draw straight lines through all unmarked rows and marked columns.

6. Revise the cost matrix as follows:

i) Find the smallest element not covered by any of the lines.

ii) Subtract this from all the uncovered elements and add it to the elements at the intersection of the two lines.

iii) Other elements covered by the lines remain unchanged.

7. Repeat the procedure until an optimum solution is attained.

Hungarian Method Example

	Machine				
J		1	2	3	
O	1	5	7	9	1
B	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Hungarian Method Example

	Machine				
J		1	2	3	
O	1	5	7	9	1
B	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Step 1: Select the smallest value in each row.

	Machine				
J		1	2	3	
O	1	5	7	9	1
B	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Hungarian Method Example

	Machine				
J		1	2	3	
O	1	5	7	9	1
B	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Step 1: Select the smallest value in each row.
Subtract this value from each value in that row

	Machine				
J		1	2	3	
O	1	0	2	4	1
B	2	4	0	2	1
	3	2	0	3	1
		1	1	1	

10-10

5-5

Hungarian Method Example

Step 2: Do the same for the columns that do not have any zero value.

	Machine				
J		1	2	3	
O	1	0	2	4	1
B	2	4	0	2	1
	3	2	0	3	1
		1	1	1	

Hungarian Method Example

smallest value in each column

	Machine				
J O B		1	2	3	
	1	0	2	4	1
	2	4	0	2	1
	3	2	0	3	1
		1	1	1	

	Machine				
J O B		1	2	3	
	1	0	2	2	1
	2	4	0	0	1
	3	2	0	1	1
		1	1	1	

Hungarian Method Example

	Machine				
J		1	2	3	
O	1	0	2	2	1
B	2	4	0	0	1
	3	2	0	1	1
		1	1	1	

Step 3: Assignments are made at zero values.

- Therefore, we assign job 1 to machine 1; job 2 to machine 3, and job 3 to machine 2.
- Total cost is 5+12+13 = 30.
- It is not always possible to obtain a feasible assignment as in here.

		Machine			
		1	2	3	
Job	1	5	7	9	1
	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Example 2

A department head has four subordinates and four tasks have to be performed. Subordinates differ in efficiency and tasks differ in their intrinsic difficulty. Time each man would take to perform each task is given in the effectiveness matrix. How the tasks should be allocated to each person so as to minimize the total man-hours?

Subordinates					
T		1	2	3	4
A	A	8	26	17	11
S	B	13	28	4	26
K	C	38	19	18	15
S	D	19	26	24	10

Subordinates					
T		1	2	3	4
A	A	8	26	17	11
S	B	13	28	4	26
K	C	38	19	18	15
S	D	19	26	24	10

Solution

Row Reduced Matrix

Subordinates					
T		1	2	3	4
A	A	0	18	9	3
S	B	9	24	0	22
K	C	23	4	3	0
S	D	9	16	14	0

Solution

Subordinates					
T		1	2	3	4
A	A	0	18	9	3
S	B	9	24	0	22
K	C	23	4	3	0
S	D	9	16	14	0

Column Reduced Matrix

Subordinates					
T		1	2	3	4
A	A	0	14	9	3
S	B	9	20	0	22
K	C	23	0	3	0
S	D	9	12	14	0

Solution

Subordinates					
T		1	2	3	4
A	A	0	14	9	3
S	B	9	20	0	22
K	C	23	0	3	0
S	D	9	12	14	0

Zero assignment

Subordinates					
T		1	2	3	4
A	A	0	14	9	3
S	B	9	20	0	22
K	C	23	0	3	0
S	D	9	12	14	0

Optimal assignment $A \to 1, \mathrm{B} \to 3, C \to 2, D \to 4$

Solution

Subordinates					
T		1	2	3	4
A	A	8	26	17	11
S	B	13	28	4	26
K	C	38	19	18	15
S	D	19	26	24	10

Zero assignment

Subordinates					
T		1	2	3	4
A	A	0	14	9	3
S	B	9	20	0	22
K	C	23	0	3	0
S	D	9	12	14	0

Optimal assignment $A \to 1, \mathrm{B} \to 3, C \to 2, D \to 4$

$$\text{Total man-hours} = 8 + 4 + 19 + 10 = 41 \text{ hours}$$

Hungarian Method Example 2

	1	2	3	4
1	***1***	4	6	3
2	9	***7***	10	9
3	***4***	5	11	7
4	8	7	8	***5***

	1	2	**3**	4
1	0	3	**5**	2
2	2	0	**3**	2
3	0	1	**7**	3
4	3	2	**3**	0

	1	2	3	4
1	***0***	3	2	2
2	2	***0***	0	2
3	0	1	4	3
4	3	2	***0***	0

Hungarian Method Example 2

- A feasible assignment is not possible at this moment.
- In such a case, The procedure is to draw a ***minimum*** number of ***lines*** through some of the rows and columns, ***Such that all zero values are crossed out***.

Hungarian Method Example 2

	1	2	3	4
1	0	3	2	2
2	2	0	0	2
3	0	1	4	3
4	3	2	0	0

The next step is to select the smallest uncrossed out element. This element is ***subtracted from every uncrossed out element*** and ***added to every element at the intersection*** of two lines.

	1	2	3	4
1	***0***	2	1	1
2	3	0	***0***	2
3	0	***0***	3	2
4	4	2	0	***0***

Hungarian Method Example 2

- We can now easily assign to the zero values. Solution is to assign (1 to 1), (2 to 3), (3 to 2) and (4 to 4).
- If drawing lines do not provide an easy solution, then we should perform the task of drwaing lines one more time.
- Actually, we should continue drawing lines until a feasible assignment is possible.

Unbalanced assignment problem

Example: Solve the following unbalanced assignment problem of minimizing the total time for performing all the jobs.

	Jobs					
W		**1**	**2**	**3**	**4**	**5**
o	**A**	5	2	4	2	5
r	**B**	2	4	7	6	6
k	**C**	6	7	5	8	7
e	**D**	5	2	3	3	4
r	**E**	8	3	7	8	6
s	**F**	3	6	3	5	7

Example: Solve the following unbalanced assignment problem of minimizing the total time for performing all the jobs.

	Jobs					
W		**1**	**2**	**3**	**4**	**5**
o	**A**	5	2	4	2	5
r	**B**	2	4	7	6	6
k	**C**	6	7	5	8	7
e	**D**	5	2	3	3	4
r	**E**	8	3	7	8	6
s	**F**	3	6	3	5	7

Solution

In this problem, the number of jobs is less than the number of workers so we have to introduce a dummy job with zero duration. The revised assignment problem is as follows:

	Jobs						
W		1	2	3	4	5	**0**
o	A	5	2	4	2	5	**0**
r	B	2	4	7	6	6	**0**
k	C	6	7	5	8	7	**0**
e	D	5	2	3	3	4	**0**
r	E	8	3	7	8	6	**0**
s	F	3	6	3	5	7	**0**

Now the problem becomes balanced one since the number of workers is equal to the number jobs. So that the problem can be solved using Hungarian Method.

Step 1: The cost Table

Step 1: Select the smallest value in each row.

Jobs							
W		1	2	3	4	5	**0**
o	A	5	2	4	2	5	**0**
r	B	2	4	7	6	6	**0**
k	C	6	7	5	8	7	**0**
e	D	5	2	3	3	4	**0**
r	E	8	3	7	8	6	**0**
s	F	3	6	3	5	7	**0**

Subtract this value from each value in that row

Jobs							
W		1	2	3	4	5	**0**
o	A	5	2	4	2	5	**0**
r	B	2	4	7	6	6	**0**
k	C	6	7	5	8	7	**0**
e	D	5	2	3	3	4	**0**
r	E	8	3	7	8	6	**0**
s	F	3	6	3	5	7	**0**

Does not change

Step 2: Select the smallest value in each columns

	Jobs						
W		1	2	3	4	5	6
o	A	5	2	4	2	5	0
r	B	2	4	7	6	6	0
k	C	6	7	5	8	7	0
e	D	5	2	3	3	4	0
r	E	8	3	7	8	6	0
s	F	3	6	3	5	7	0

Subtract this value from each value in that row

	Jobs						
W		1	2	3	4	5	6
o	A	3	0	1	0	1	0
r	B	0	2	4	4	2	0
k	C	4	5	2	6	3	0
e	D	3	0	0	1	0	0
r	E	6	1	4	6	2	0
s	F	1	4	0	3	3	0

Step 4: Determine an Assignment By using the Hungarian method, the assignment is made as follows:

	Jobs						
W		1	2	3	4	5	6
o	A	3	~~0~~	1	0	1	~~0~~
r	B	0	2	4	4	2	0
k	C	4	5	2	6	3	0
e	D	3	0	0	1	0	0
r	E	6	1	4	6	2	0
s	F	1	4	0	3	3	0

Step 4: Determine an Assignment By using the Hungarian method, the assignment is made as follows:

	Jobs						
W		1	2	3	4	5	6
o	A	3	~~0~~	1	0	1	~~0~~
r	B	0	2	4	4	2	~~0~~
k	C	4	5	2	6	3	0
e	D	3	0	0	1	0	0
r	E	6	1	4	6	2	0
s	F	1	4	0	3	3	0

Step 4: Determine an Assignment By using the Hungarian method, the assignment is made as follows:

	Jobs						
W		1	2	3	4	5	6
o	A	3	~~0~~	1	0	1	~~0~~
r	B	0	2	4	4	2	~~0~~
k	C	4	5	2	6	3	0
e	D	3	~~0~~	~~0~~	1	0	~~0~~
r	E	6	1	4	6	2	~~0~~
s	F	1	4	0	3	3	~~0~~

Step 4: Determine an Assignment By using the Hungarian method, the assignment is made as follows:

	Jobs						
W		1	2	3	4	5	6
o	A	3	0	1	0	1	0
r	B	0	2	4	4	2	0
k	C	4	5	2	6	3	0
e	D	3	0	0	1	0	0
r	E	6	1	4	6	2	0
s	F	1	4	0	3	3	0

Step 5:

The solution obtained in Step 4 is not optimal. Because we were able to make five assignments when six were required.

Step 6:

Cover all the zeros of the table shown in the Step 4 with five lines (since already we made five assignments). Check row E since it has no assignment. Note that row B has a zero in column 6, therefore check column 6. Then we check row C since it has a zero in column 6. Note that no other rows and columns are checked. Now we may draw five lines through unchecked rows (row A, B, D and F) and the checked column (column 6). This is shown in the table given below:

	Jobs						
W		1	2	3	4	5	6
o	A	3	0	1	0	1	0
r	B	0	2	4	4	2	0
k	C	4	5	2	6	3	0
e	D	3	0	0	1	0	0
r	E	6	1	4	6	2	0
s	F	1	4	0	3	3	0

Step 7:

Develop the new revised table. Examine those elements that are not covered by a line in the table given in Step 6. Take the smallest element, in this case the smallest element is 1.

Subtract this smallest element from the uncovered cells and add 1 to elements (A6, B6, D6 and F6) that lie at the intersection of two lines. Finally, we get the new revised cost table, which is shown below:

	Jobs						
W		1	2	3	4	5	**6**
o	A	3	0	1	0	1	**0**
r	B	0	2	4	4	2	**0**
k	C	4	5	2	6	3	**0**
e	D	3	0	0	1	0	**0**
r	E	6	1	4	6	2	**0**
s	F	1	4	0	3	3	**0**

	Jobs						
W		1	2	3	4	5	**6**
o	A	3	0	1	0	1	**0**
r	B	0	2	4	4	2	**0**
k	C	3	4	1	5	2	**0**
e	D	3	0	0	1	0	**0**
r	E	5	0	3	5	1	**0**
s	F	1	4	0	3	3	**0**

Step 8: Now, we go to Step 4 and repeat the procedure until we arrive at an optimal solution (assignment).

Step 9: Determine an assignment

	Jobs						
W		1	2	3	4	5	**6**
o	A	3	0	1	0	1	**0**
r	B	0	2	4	4	2	**0**
k	C	3	4	1	5	2	**0**
e	D	3	0	0	1	0	**0**
r	E	5	0	3	5	1	**0**
s	F	1	4	0	3	3	**0**

Step 8: Now, we go to Step 4 and repeat the procedure until we arrive at an optimal solution (assignment).

Step 9: Determine an assignment

Workers	Jobs						
W		1	2	3	4	5	6
o	A	3	0	1	0	1	0
r	B	0	2	4	4	2	0
k	C	3	4	1	5	2	0
e	D	3	0	0	1	0	0
r	E	5	0	3	5	1	0
s	F	1	4	0	3	3	0

Workers	Jobs					
W		**1**	**2**	**3**	**4**	**5**
o	**A**	5	2	4	2	5
r	**B**	2	4	7	6	6
k	**C**	6	7	5	8	7
e	**D**	5	2	3	3	4
r	**E**	8	3	7	8	6
s	**F**	3	6	3	5	7

A4 + B1 + D5 + E2 + F3 = 2 + 2 + 3 + 3 + 3 = 14

Since the number of assignments equal to the number of rows (columns), the assignment shown in the above tale is optimal. Thus, the worker A is assigned to Job 4, worker B is assigned to job 1, worker C is assigned to job 6, worker D is assigned to job 5, worker E is assigned to job 2, and worker F is assigned to job 3. Since the Job 6 is dummy so that worker C can’t be assigned.

The total minimum time is:

A4 + B1 + D5 + E2 + F3 = 2 + 2 + 4 + 3 + 3 = 14